Project Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	40
Problem Understanding	10					
Analysis	15					
Implementation	10					
Task Efficiency	5					
					Total obtained mark	
Comments						

Semester: Spring 2025

Student Name: Arif Hosen **Student ID: 251-35-187**

Student Name: Nisa Hannan **Student ID: 251-35-104**

Student Name: Antor Biswas **Student ID: 251-35-365**

Batch: 44 **Section :H(1)** **Course Code: SE133**

Course Name: Software Development Capstone Project

Course Teacher Name: Jaodun Muntasir

Designation: Lecturer

Submission Date: 04/12/25

Table of Contents

1. **Introduction ……**

1.1 System Overview ……

1.2 Why we developed this system?.....

2. **Stakeholders……**

2.1 What is a stakeholder?.....

2.2 Who are the stakeholders for this system?.....

3. **Requirement Analysis…….**

3.1 What is Requirement Analysis?.....

3.2 Functional Requirements…….

3.3 Non-Functional Requirements……

4. **Use Case Diagram…….**
5. **User Manual……**

5.1 Feature Description for Logged Customers……

5.2 Feature Description for Logged Staff (Administrators & Technicians) …….

6. **Project Plan……**
7. **Testing & Analysis……**
8. **Conclusion……**
9. **Future Work…….**

1. Introduction

In a busy automotive repair environment, garages must manage numerous vehicles, parts, jobs, and bills simultaneously. Manual processes lead to delays, lost records, errors in billing, and poor customer experience.

The Garage Service Management System is a console-based C program designed to automate core garage workflows: vehicle intake, service stage tracking (inspection → diagnosis → repair → completion), parts usage tracking, labor & materials billing, and invoice generation. The system persists data using file-handling and provides role-based access for customers and garage staff.

1.1 System Overview

- Logs incoming vehicles with unique Service IDs.
- Records customer details and vehicle information.
- Tracks service stages and allows staff notes at each stage.
- Records spare parts usage and inventory updates.
- Calculates labor, parts and total charges and generates invoices.
- Maintains service history for future diagnostics.

1.2 Why we developed this system?

- Reduce manual error and paperwork.
- Improve traceability and customer transparency.
- Provide faster service logging and billing.
- Maintain persistent service history to assist repeat repairs.

2. Stakeholders

2.1 What is Stakeholders?

Stakeholders in a software system are individuals, groups, or organizations that are impacted by the system's development, use, or existence. They have a vested interest in the system's success and can influence its requirements, design, and functionality.

2.2 Stakeholders for this system

- **Non-logged Customers:** Can view general information and must register to request services.
- **Logged Customers:** Can request service, view status and service history, and receive invoices.
- **Garage Staff (Administrators & Technicians):** Manage vehicle intake, update service status, record parts used, compute charges, and generate invoices.

3. Requirement Analysis:

3.1 What is Requirement Analysis?

Requirements Analysis is a fundamental stage in software development that focuses on determining the needs, expectations, and conditions that a new or modified product or system should fulfill. This process considers the often-conflicting requirements of various stakeholders involved in the project.

3.2 Functional Requirements for this system

We will be using pharmacist and an overall general customer in regards of logged in or not.

FR01	Login
Description	Registered customers, technicians/administrators can log in.
Stakeholders	Customer, Staff

FR02	Log New Vehicle for Service
Description	Staff registers vehicle, owner details, reported issues and generates a unique Service ID.
Stakeholders	Customers

FR03	Update Service Status
Description	Staff updates stages: Inspection → Diagnosis → Repair → Completed.
Stakeholders	Staff

FR04	Add Service Notes
Description	Staff can add textual notes per stage.
Stakeholders	Staff

FR05	Add Spare Parts Used
Description	Staff records parts with quantity and cost.
Stakeholders	Staff

FR06	Manage Parts Inventory
Description	System updates inventory when parts used.
Stakeholders	Staff

FR07	Calculate Charges
Description	System computes labor cost (hourly) + parts cost and taxes if applicable.
Stakeholders	Garage Manager, Costumer.

FR08	Generate Invoice
Description	System prints/saves a detailed invoice with items, labor, subtotal, taxes, total.
Staeholders	Garage manager

FR9	View Service History
Description	Customers and staff can view previous services by vehicle or customer ID.
Stakeholders	Staff, Customers.

FR10	Search Service Records
Description	Staff can search by Service ID, vehicle number, or customer name.
Stakeholders	Staff

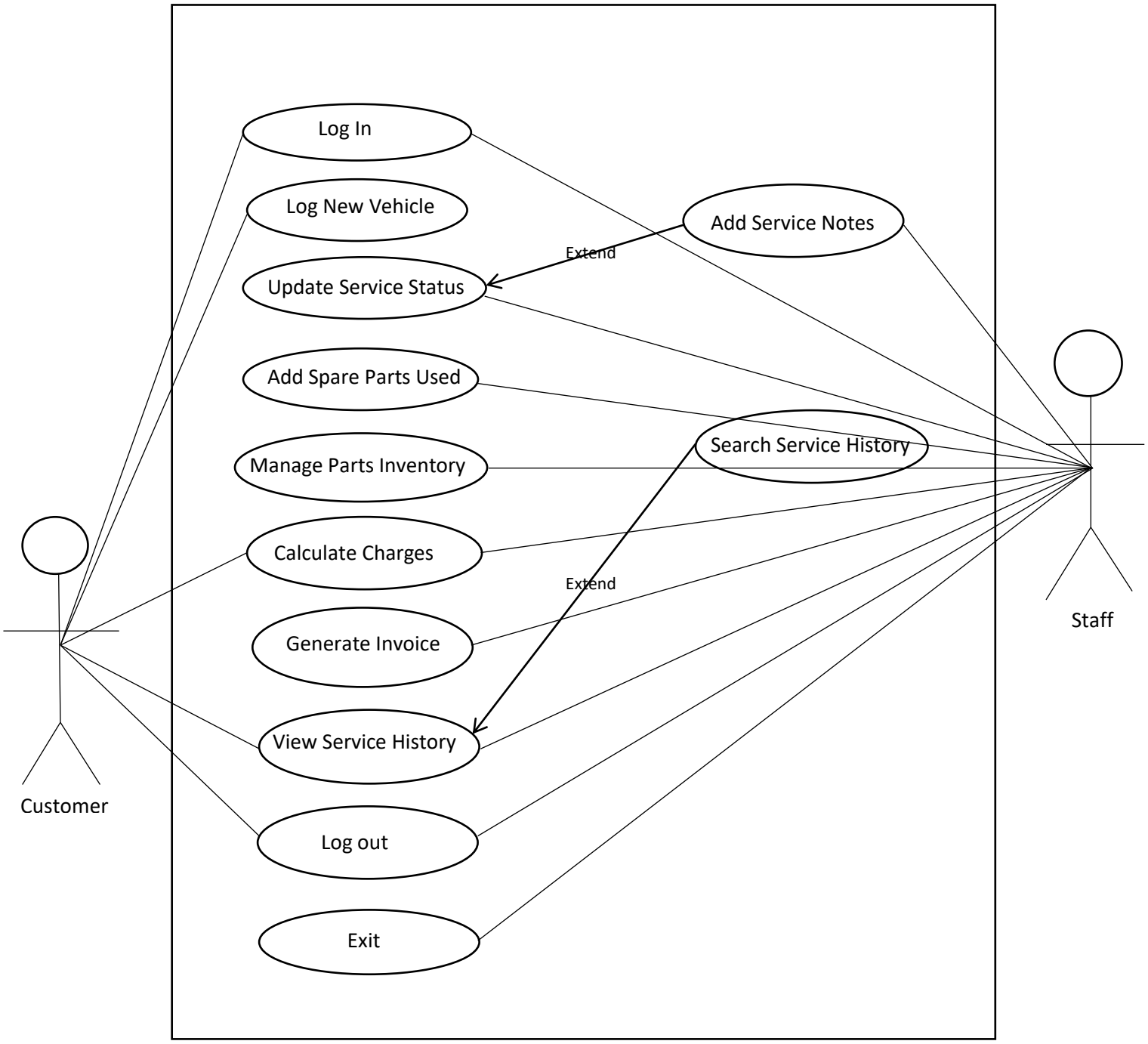
FR11	Logout
Description	User can log out.
Stakeholders	Customers

FR12	Exit
Description	Exit the application.
Stakeholders	Customers.

3.3 Non-Functional Requirements

- **Maintainability:** Modular C code (functions & structures) for ease of modification.
- **Availability:** Runs as a local standalone program.
- **Accuracy:** Correct calculations and consistent file updates.
- **Security:** Simple authentication for staff and customers.

4. Use Case Diagram

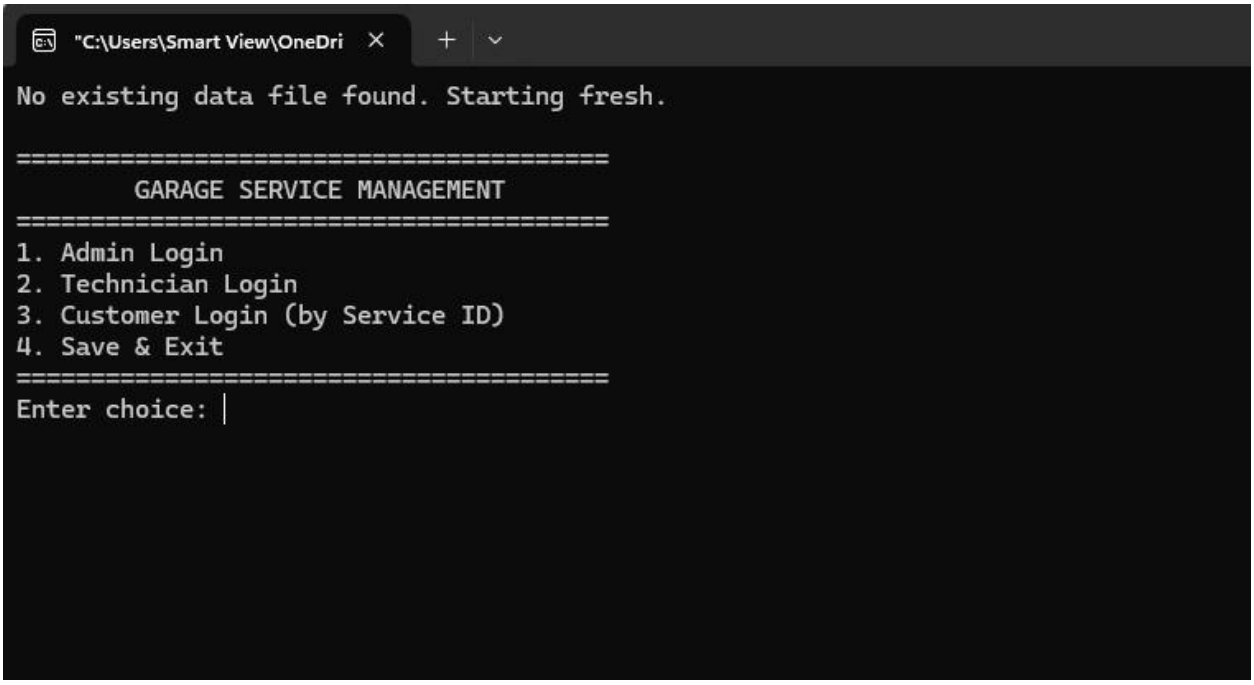


5. User Manual

1. Start:

Main menu:

This is the main menu of a Garage Service Management System. It provides options to Login as an admin, technician and customers, Save & Exit. Users select a choice by entering the corresponding number.



2.1 Admin Login:

After entering choice, the system asks for the password for its safety and protection.

```
"C:\Users\Smart View\OneDri

No existing data file found. Starting fresh.

========================================
        GARAGE SERVICE MANAGEMENT
========================================
1. Admin Login
2. Technician Login
3. Customer Login (by Service ID)
4. Save & Exit
========================================
Enter choice: 1
Enter Admin password: admin123
```

2.1.1 Add New Vehicle (optionally via staff):

This system allows the user to input information about a new customer & his vehicle details into the system's database.

```
--- Add New Vehicle (Service ID: 1) ---
Enter Customer Name: Antor Biswas
Enter Customer Phone: 01794444444
Enter Vehicle Name/Type: Cycle
Enter number of reported issues (0-6): 2
Issue 1: flat tire
Issue 2: chain
Added successfully. Service ID = 1
```

2.1.2 Search Info:

The "Search Info" function lets the admin quickly find a vehicle/service record.

```
--- ADMIN MENU ---
1. AddNewVehicle
2. SearchInfo
3. UpdateInfo
4. Delete
5. ServiceStatusz
6. AddNotes
7. BillCount
8. Display
9. PartsUsed
10. Back to Main Menu
Enter choice: 2

--- SearchInfo ---
1. Search by Service ID
2. Search by Customer Name (substring)
3. Search by Vehicle Name (substring)
Enter choice: 2
Enter customer name substring: Sadhin
ID:2 | Customer:Sadhin | Vehicle:Bike | Status:0
ID:4 | Customer:Sadhin | Vehicle:cycle | Status:0
```

2.1.3 Update Info:

The Update Info option lets the admin modify existing vehicle or customer service details in the system.

```
--- ADMIN MENU ---
1. AddNewVehicle
2. SearchInfo
3. UpdateInfo
4. Delete
5. ServiceStatusz
6. AddNotes
7. BillCount
8. Display
9. PartsUsed
10. Back to Main Menu
Enter choice: 3

--- UpdateInfo ---
Enter Service ID to update: 3
Current Customer Name: Upoma Nisa
Enter new name (or press Enter to keep): Upoma
Current Customer Phone: 01740000000
Enter new phone (or press Enter to keep): 01794444444
Current Vehicle Name: car
Enter new vehicle name (or press Enter to keep): car
Do you want to change reported issues? (y/n): n
Do you want to change labor charge? (y/n): n
Record updated and saved.
```

2.1.4 Delete:

The delete option removes unwanted or incorrect vehicle service records using the unique vehicle ID.

```
--- ADMIN MENU ---
1. AddNewVehicle
2. SearchInfo
3. UpdateInfo
4. Delete
5. ServiceStatusz
6. AddNotes
7. BillCount
8. Display
9. PartsUsed
10. Back to Main Menu
Enter choice: 4

--- Delete ---
1. Delete single vehicle record by Service ID
2. Delete all records of a customer (by exact customer name)
Enter choice: 1
Enter Service ID to delete: 2
Delete record: ID:2 | Customer:Sadhin | Vehicle:Bike ? (y/n): y
Record deleted successfully.
```

2.1.5 Service Status:

The Service Status function updates the progress stage of a vehicle's service. After entering a service ID, it shows the current status and provides options like Inspection, Diagnosis, Repairing, or Completed.

```
--- ADMIN MENU ---
1. AddNewVehicle
2. SearchInfo
3. UpdateInfo
4. Delete
5. ServiceStatusz
6. AddNotes
7. BillCount
8. Display
9. PartsUsed
10. Back to Main Menu
Enter choice: 5

--- ServiceStatusz ---
Enter Service ID: 2
Current status: 0
Choose new status:
1. Inspection
2. Diagnosis
3. Repairing
4. Completed
Enter status number: 3
Status updated.
```

2.1.6 Add Notes:

The Add Notes function lets the admin record additional details about a vehicle□s service, such as observations, mechanic comments, or special instructions.

```
--- ADMIN MENU ---
1. AddNewVehicle
2. SearchInfo
3. UpdateInfo
4. Delete
5. ServiceStatusz
6. AddNotes
7. BillCount
8. Display
9. PartsUsed
10. Back to Main Menu
Enter choice: 6

--- AddNotes ---
Enter Service ID: 1
Enter note to add: You have to wate another 2 days
Note saved.
```

2.1.7 Bill Count:

The Bill Count function calculates the total service cost for a vehicle by adding labor charges, spare parts used, and any additional fees.

```
--- ADMIN MENU ---
1. AddNewVehicle
2. SearchInfo
3. UpdateInfo
4. Delete
5. ServiceStatusz
6. AddNotes
7. BillCount
8. Display
9. PartsUsed
10. Back to Main Menu
Enter choice: 7

--- BillCount ---
Enter Service ID: 1
Current labor charge: 0.00
Enter labor charge (or press Enter to keep): 100
Do you want to add a part? (y/n): y
Enter part name: chain
Enter quantity: 1
Enter unit price: 150
Billing updated. Parts: 150.00 | Labor: 100.00 | Total: 250.00
```

2.1.8 Display:

The Display function shows all the service records stored in the system in a clean, table-like format.

```
--- ADMIN MENU ---
1. AddNewVehicle
2. SearchInfo
3. UpdateInfo
4. Delete
5. ServiceStatusz
6. AddNotes
7. BillCount
8. Display
9. PartsUsed
10. Back to Main Menu
Enter choice: 8

--- All Service Records (4) ---

ServiceID  Customer Name            Vehicle Name          Status        Total Bill
--------------------------------------------------------------------------------
1          Antor Biswas            Cycle                 Intake            300.00
2          Sadhin                  Bike                  Repairing           0.00
3          Upoma                   car                   Completed         500.00
4          Sadhin                  cycle                 Intake              0.00
--------------------------------------------------------------------------------
```

2.1.9 Parts Used:

The Parts Used function shows information about the spare parts used during vehicle servicing.

```
--- ADMIN MENU ---
1. AddNewVehicle
2. SearchInfo
3. UpdateInfo
4. Delete
5. ServiceStatusz
6. AddNotes
7. BillCount
8. Display
9. PartsUsed
10. Back to Main Menu
Enter choice: 9

--- PartsUsed ---
1. View parts for one Service ID
2. View total parts used across all records
Enter choice: 2
Total parts used across all records: 1
Total parts cost across all records: 150.00
```

2.2 Technician Login:

It provides options to check service status, add notes, review or update parts used, and search for information, making the workflow organized and efficient.

```
--- TECHNICIAN MENU ---
1. ServiceStatusz
2. AddNotes
3. PartsUsed (view / add parts via BillCount by admin)
4. SearchInfo
5. Back to Main Menu
Enter choice: |
```

2.2.1 Service Status:

The Service Status section allows a technician to view and update the progress of a service request. It shows the current status (like *Intake*, *Inspection*, *Diagnosis*, *Repairing*, *Completed*) and lets the technician choose a new status by entering the corresponding number, helping keep the workflow clear and up-to-date.

```
--- TECHNICIAN MENU ---
1. ServiceStatusz
2. AddNotes
3. PartsUsed (view / add parts via BillCount by admin)
4. SearchInfo
5. Back to Main Menu
Enter choice: 1

--- ServiceStatusz ---
Enter Service ID: 3
Current status: Completed
Choose new status:
0. Intake
1. Inspection
2. Diagnosis
3. Repairing
4. Completed
Enter status number: 4
Status updated to: Completed
```

2.2.2 Search Info:

The Search Info feature lets technicians quickly look up detailed service records.
They can search by Service ID**,** Customer Name**, or** Vehicle Name, and the system displays full service details such as customer info, issues, status, and total charges—making it easy to track and review any service case.

```
--- TECHNICIAN MENU ---
1. ServiceStatusz
2. AddNotes
3. PartsUsed (view / add parts via BillCount by admin)
4. SearchInfo
5. Back to Main Menu
Enter choice: 4

--- SearchInfo ---
1. Search by Service ID
2. Search by Customer Name (substring)
3. Search by Vehicle Name (substring)
Enter choice: 1
Enter Service ID: 3

--- Service Full Details ---
Service ID : 3
Customer   : Upoma
Phone      : 01794444444
Vehicle    : car
Status     : Completed
Issues:
  - flat tire
  - oil leak
  - head light
Parts total: 0.00
Labor charge: 500.00
Total charge: 500.00
```

2.3 Customer Login:

```
--- CUSTOMER MENU (Service ID: 3) ---
1. View Service Status (summary)
2. View Invoice (if completed) - full details
3. Back to Main Menu
Enter choice: |
```

2.3.1 View Service Status:

The View Service Status option allows customers to quickly check the current progress of their vehicle's service. It shows a simple summary including the service ID, customer name, vehicle type, status stage, and total bill (if available). This feature keeps customers informed without needing staff assistance.

```
--- CUSTOMER MENU (Service ID: 3) ---
1. View Service Status (summary)
2. View Invoice (if completed) - full details
3. Back to Main Menu
Enter choice: 1

Service ID: 3
Customer: Upoma
Vehicle: car
Status: Completed
Total Bill: 500.00
```

2.3.2 View Invoice:

The View Invoice option lets customers see the complete billing details once their service is finished. It displays the customer's information, vehicle type, service status, list of reported issues, parts cost, labor charge, and the final payable amount. This ensures full transparency and provides a clear breakdown of all service charges.

```
--- CUSTOMER MENU (Service ID: 3) ---
1. View Service Status (summary)
2. View Invoice (if completed) - full details
3. Back to Main Menu
Enter choice: 2

--- INVOICE (Service ID: 3) ---
Customer: Upoma | Phone: 01794444444
Vehicle: car
Status: Completed
Issues:
- flat tire
- oil leak
- head light
Parts total: 0.00
Labor charge: 500.00
TOTAL PAYABLE: 500.00
```

6. Project Plan

Timeline

- Requirement gathering: Week 1–2
- System design: Week 2–3
- Core development: Week 3–6
- Testing & feedback: Week 6–8
- Deployment & training: Week 8–10

Deliverables

- Complete C program with source code
- Project report (this document)
- PPTX presentation
- Test cases and test report

Resources

- Development: C & GCC Compiler
- Team: Developers, testers, documenters
- Hardware: PCs, printers

7. Testing & Analysis

7.1 Functional Testing

Customer: login/registration, view history, view invoices.

Staff: log vehicle, update stages, add parts, generate invoices, search records.

7.2 Non-Functional Testing

- Performance: menu response and file operations under load.
- Security: login checks.
- Usability: clarity of menu options and messages.

7.3 Defect Tracking

Log defects in a simple text log: ID, description, reproduce steps, status.

---

8. Conclusion

The Garage Service Management System digitizes garage operations, ensuring accurate tracking of repairs and billing while improving transparency and customer satisfaction.

Future Work:

- SMS/email notifications for service status updates.
- Web-based frontend and remote booking.
- Barcode-based parts scanning and better inventory analytics.